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Key Points:

- $^{239+240}\text{Pu}$ activity is a robust tracer for describing sediment sources and fluvial mixtures in a rapidly urbanizing landscape
- Frequentist and Bayesian models were consistent in estimating increased bank erosion with increasing urban land use
- Sediment sourcing is dependent on discharge in rural watersheds whereas it is independent of discharge in urban watersheds

Supporting Information:

Supporting Information may be found in the online version of this article.

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Plutonium Isotopes: An Effective Tool for Fluvial Sediment Sourcing in Urbanized Catchments

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Abstract Sediment management is currently limited by uncertainties in the applicability of existing radionuclide tracers and the effects of urbanization. Here, we use plutonium isotopes ($^{239+240}\text{Pu}$) from weapons-testing fallout to trace sediment transport across five watersheds in an urbanizing landscape in Kansas, USA. Historic flooding in the region provided an opportunity to assess extreme connectivity of sediment sources. $^{239+240}\text{Pu}$ activity of transported sediment decreased as catchments urbanized, indicating a greater contribution from subsurface bank sediment; Bayesian modeling predicted 50% (8%–80%) bank sourcing in the most rural watershed, which rose to 93% (73%–100%) in the most urban catchment. $^{239+240}\text{Pu}$ activity provides explanatory information on the superposition of sediment sources, which is beyond that given by traditional organic and geochemical tracers that primarily infer vegetative and geologic sourcing, respectively. Our study demonstrates the utility of $^{239+240}\text{Pu}$ as a sediment tracer for managing erosion under anthropogenic change.

Plain Language Summary Sediment is one of the most common contaminants in rivers around the globe. The susceptibility of soil to erosion increases as humans convert natural landscapes into rural and urban systems. Uncertainties remain regarding how urbanization alters sediment pathways and what tools are appropriate for quantifying this alteration. Radioactive isotopes, generated from nuclear weapons testing in the 1950s, have proven useful in this regard, although the most prominent tracer, for example, cesium, is losing some utility due to a short half-life and extensive radioactive decay. We suggest that plutonium isotopes, which have longer half-lives, are a viable option for tracing sediment in modern landscapes. Plutonium results suggest that in rural basins the source of sediment transitions from upland soil during low-flows to bank material at high-flows. On the other hand, urban streams always deliver bank sediment, regardless of storm intensity. Our study demonstrates the utility of plutonium as an alternate sediment tracer and highlights how urbanization changes the pathways and mechanisms of fluvial sediment transport.

1. Introduction

Much of the Earth's land surface is experiencing anthropogenically driven land use change and erosion at unprecedented rates (Owens, 2020; Russell et al., 2017; Shi et al., 2021). The rapid conversion of natural and rural landscapes into urban systems has diminished agricultural productivity, increased sediment pollution, and has counteracted urban sustainability (Chang et al., 2020; Zarnaghsh & Husic, 2021). The global urban population is expected to double by 2050 (Bettencourt et al., 2018), which will be met with increasing urban development as well as agricultural activity to satisfy population demands.

Managing erosion and tracing sediment back to its origin is complicated by uncertainties in the connectivity of pathways, flow-dependence of source-zone activation, and sensitivity of tracers to successfully identify sources (Husic et al., 2020; Michalek et al., 2021; Owens, 2020). The most widely employed radioisotope, Cesium-137 (^{137}Cs , $t_{1/2} = 30.2$ years), was deposited on surficial soil in large quantities during nuclear weapons testing in the 1960s (Ritchie & McHenry, 1990). However, in the intervening decades, ^{137}Cs has reduced to just ~25% of its original levels, rendering its application more challenging as time progresses limiting its utility as a tracer with conventional techniques (Alewell et al., 2017; Collins et al., 2020) although recent analytical advancements are allowing sensitive detection of ^{137}Cs (Povinec, 2018). A recent study of radionuclides in sediment tracing calls on the community to identify new tracers for soil redistribution as depleted tracers lose some viability (Alewell et al., 2017).

Plutonium ($^{239+240}\text{Pu}$) has garnered significant attention as an alternative tracer of erosion as it was globally deposited in the same manner as ^{137}Cs , but consists of two major long-lived isotopes (^{239}Pu , $t_{1/2} = 24,110$ years;

^{240}Pu , $t_{1/2} = 6,561$ years, Alewell et al., 2017; Ketterer et al., 2004; Meusbürger et al., 2018). Until recent advancements in inductively coupled plasma spectrometry (ICP-MS, Ketterer & Szechenyi, 2008), $^{239+240}\text{Pu}$ analysis required alpha-particle spectrometry (Schimmack et al., 2001), making it inaccessible and cost-prohibitive. $^{239+240}\text{Pu}$ has proven successful in related applications such as sediment chronology (Evrard et al., 2014; Meusbürger et al., 2016; Tims et al., 2010) and redistribution (Arata et al., 2016; Calitri et al., 2019; Johansen et al., 2019; Smith et al., 2012) though it has not yet been tested in tracing fluvial sediment transport.

Sediment fingerprinting is a management tool for tracking the origin, movement, and fate of eroded soil (Collins et al., 2017; Davis & Fox, 2009). The ability of a fingerprinting model to accurately determine source contributions is a function of adequate selection of tracers, identification of sources, and choice of model structure (Collins et al., 2017). To the last point, two modeling frameworks prevail in sediment fingerprinting studies: frequentist and Bayesian (Cooper et al., 2014). Frequentist approaches make no assumptions about the structure of modeled residuals whereas Bayesian methods can account for end-member covariance, a priori distributions, and nonconservative tracer behavior (Arendt et al., 2015; Cooper et al., 2014). The integration of multiple structures into an analysis can be useful for ascertaining the sensitivity of model choice on apportionment estimates; however, this type of sensitivity analysis has not been performed on plutonium sediment fingerprinting.

The objective of this study was to assess the utility of plutonium isotopes as tracers of fluvial sediment in urbanized catchments. Over the course of a year of historic flooding, we collected fortnightly fluvial suspended sediment samples from five rivers along an urbanization gradient in Kansas, USA. We conducted ICP-MS-based measurements of $^{239+240}\text{Pu}$ activities and constructed frequentist and Bayesian mixing models to apportion sediment to surface (upland) and subsurface (bank) sources. Lastly, we provide a conceptual model of the drivers of $^{239+240}\text{Pu}$ activity variability as a function of geochemical soil properties, storm intensity, and land use.

2. Study Site and Materials

To meet the objectives of this project, urbanizing Johnson County, Kansas, USA, was selected as the study site (Figure 1). The area is experiencing expansion with a population boom of 35% since 2000 from 451,000 to 610,000 (Michalek et al., 2021), compared to just 4% growth in the rest of Kansas. Johnson County has a mean slope of 3.7° and a temperate climate with mean annual (1970–2019) temperature and precipitation of $13.7 \pm 0.4^\circ\text{C}$ and $958 \pm 211 \text{ mm yr}^{-1}$, respectively. Soils in the area are primarily silt loam and silty clay loam and are underlain by interbedded limestone and shale geology (Figure S1 in Supporting Information S1). We focused on five watersheds that range considerably in their urban land use from 21% in Blue River to 89% in Indian Creek. Sediment sinks (e.g., lakes and reservoirs) exist throughout the watersheds, most prominently in Cedar Creek where they drain 35.3% of basin area. Tile-drainage of cropland varies from negligible (3.0% of Indian Creek) to prominent (18.6% of Blue River, Valayamkunnath et al., 2020). Impervious surfaces range from extensive (41.8% of Indian Creek) to minimal (7.5% of Kill Creek). Discharge data were obtained from the United States Geological Survey and precipitation data were retrieved from the Johnson County StormWatch database (StormWatch, 2020).

3. Methods

3.1. Plutonium Activity of Fluvial Sediment in Midwestern USA

3.1.1. Field Collection

To describe suspended fluvial sediment, in situ time-integrated trap samplers (Phillips et al., 2000) were installed at each watershed outlet and sampled approximately fortnightly for the 2019 calendar year (Figure S2 and Table S1 in Supporting Information S1). A range of upland sources was sampled to describe surface contribution of sediment to the transported load, including woodland, grassland, cropland, residential, and construction sites (Figure 1 and Table S2 in Supporting Information S1). At each site, $\sim 0.1 \text{ m}^2$ of surficial sediment was disturbed by a hand trowel (Juracek & Ziegler, 2009), making sure to get only the top 2 cm of sediment likely to erode and be enriched in plutonium (Alewell et al., 2017). For each upland site, five sub-samples of $\sim 20 \text{ g}$ each were collected to account for small-scale variability within a source (Abban et al., 2016). Subsurface samples were collected at 3-km intervals along the mainstems of Blue River, Mill Creek, and Indian Creek. At each location, bank samples were collected at 15, 30, and 45 cm above the water line at two transects $\sim 20 \text{ m}$ apart to capture

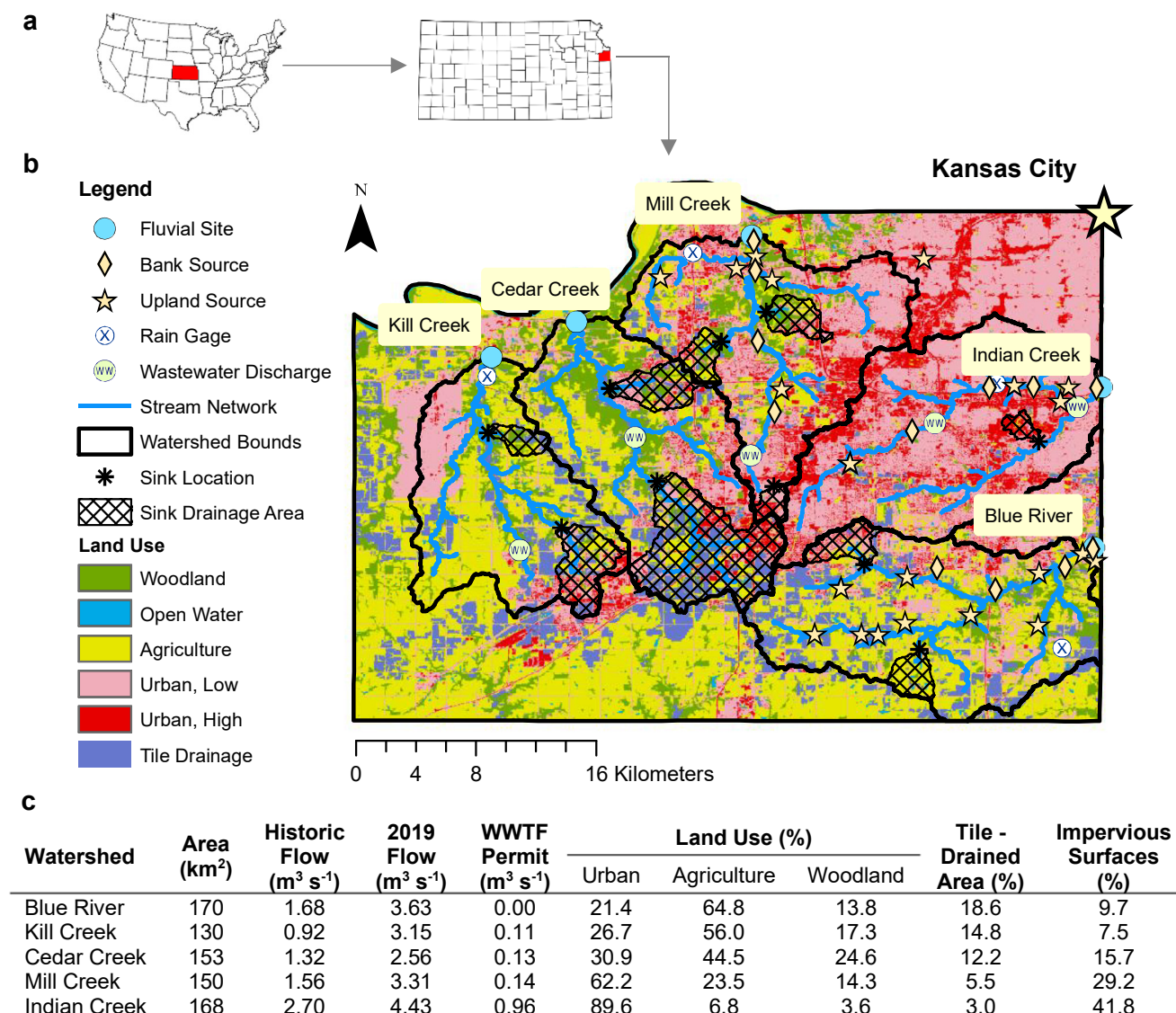


Figure 1. (a) Location of Johnson County, Kansas, USA. (b) Fluvial sediment sampling sites, source sample locations, rain gages, wastewater treatment facility discharges, stream networks, sinks (e.g., lakes, reservoirs, dams), and land use. (c) Watershed name, drainage area, mean historic flow, mean study-period flow, and permitted wastewater treatment facility (WWTF) discharge, land use, and percent tile-drained and impervious areas.

in-site variability. Source samples were only collected in Blue River, Mill Creek, and Indian Creek due to labor and time constraints, but span most land uses and soil types in the region.

3.1.2. Laboratory Analyses

Suspended and source samples were wet-sieved to retain only the fine fraction ($<63 \mu\text{m}$), freeze-dried to remove moisture, and homogenized in an amalgamator (Davis & Fox, 2009). A random subset of the transported sediment samples ($n = 24$) were analyzed for particle size distributions using a Beckman Coulter LS 13 320 to investigate particle size effects (Dong et al., 2017). $^{239+240}\text{Pu}$ activity measurements were performed at Northern Arizona University using procedures based upon Ketterer et al. (2004). Samples of 3–4 g dry mass were ashed at 450°C , spiked with ^{242}Pu tracer (49 pg/sample, prepared from NIST 4334i), and leached with 10 mL of 16 M HNO_3 for 16 hr at 80°C . The mixtures were filtered and treated with 0.1 g of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ and 0.4 g NaNO_2 to convert Pu to the Pu(IV) oxidation state. Plutonium was purified using 100 mg TEVA resin (Eichrom) columns, and was eluted using 0.05 M aqueous ammonium oxalate. The eluted Pu fractions were analyzed by ICP-MS; an APEX HF high-efficiency desolvating sample introduction system (Elemental Scientific) was used with a

Thermo X Series II quadrupole ICP-MS. The masses of ^{239}Pu and ^{240}Pu present in the sediment were determined by isotope dilution calculations and then converted into a summed $^{239+240}\text{Pu}$ activity. The method detection limit was 0.02 Bq kg^{-1} for $^{239+240}\text{Pu}$; the analysis of three control samples (IAEA-384, IAEA-367, and MAPEP 01 S8) indicated good agreement with certified/accepted activities. Samples were also analyzed for organic constituents (C, N, $\delta^{13}\text{C}$, and $\delta^{15}\text{N}$) and 20 major and minor elements (Supporting Information S1).

3.1.3. Statistical Analyses

Significant differences in the $^{239+240}\text{Pu}$ activities of source and transported sediment were assessed using the Mann Whitney U-test ($\alpha = 0.05$). The effect size of significant differences was determined using Cliff's δ where values greater than 0.50 indicate large effect size (Cliff, 1993). Pearson correlation (ρ) matrices between $^{239+240}\text{Pu}$ activity and physical, organic, and inorganic element explanatory variables were generated. Moderate correlation was inferred if $0.30 \leq |\rho| < 0.50$, whereas high correlation was inferred if $|\rho| \geq 0.50$. Principal component analysis (PCA), which reduces the dimensionality of the variable space, was used to generally describe potential drivers of $^{239+240}\text{Pu}$ activity in relation to the explanatory variables. Lastly, we investigated how the mean discharge during a storm event impacts the Plutonium activity of transported sediment. We grouped transported sediment into two categories: rural (Blue River and Kill Creek) and urban (Mill Creek and Indian Creek). Cedar Creek was excluded from the analysis due to low sample count ($n = 3$). Power and linear regressions were fit to the rural and urban data, respectively, and 95% uncertainty bounds were calculated.

3.2. Frequentist and Bayesian Source Apportionment of Fluvial Sediment

We employed two sediment fingerprinting methods to apportion transported sediment to surface and subsurface sources: (a) frequentist and (b) Bayesian. Both methods solve the following mass balance mixing model:

$$\sum_{i=1}^n \pi_i = 1 \quad (1)$$

$$\sum_{i=1}^n \pi_i y_i = P_j \quad (2)$$

where π_i is the fractional contribution from i end-member, n is the number of end members, y_i is the Plutonium activity of the i end-member, and P_j is the Plutonium activity of the j mixture. The main aspects of each method are summarized below; the reader is referred to the Supporting Information S1 for further details.

The frequentist approach solves the mixing model equations by minimizing the sum of squared residuals (Collins et al., 2020; Cooper & Krueger, 2017). Briefly, end-members are randomly sampled from collected data and the π_i values are calculated to minimize error for the random draw. For the Bayesian approach, we adapted a framework developed by Erhardt and Bedrick (2013) in MATLAB, which is similar in principle to prominent Bayesian tools (e.g., MixSIAR; Stock et al., 2018). The model follows Bayes' theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad (3)$$

where the terms in the above equation are: (a) the posterior $P(A|B)$ or the probability of an outcome A given B observation, (b) the priors $P(A)$ or the belief in observing A without any given conditions, and (c) the likelihood $P(B|A)$ given evidence $P(B)$ or the support knowledge that B provides to A (Cooper et al., 2014). The goal is to determine the likely contributions of end-members to the mixture given the observed data, that is $P(A|B)$, which the model does using an importance sampling algorithm rather than Markov chain Monte Carlo analysis (Erhardt & Bedrick, 2013). For each mixture, both models were run for 100,000 realizations and credible intervals at the 50th and 95th percentiles were constructed.

4. Results and Discussion

4.1. Plutonium Activity as a Tracer for Fluvial Sediment

During the study period (2019), our sites experienced a year of intense rainfall (1,358 mm) compared to the historical mean (958 ± 211 mm). Mean streamflows were 1.6–3.4 times the historical average and varied from $2.56 \text{ m}^3 \text{ s}^{-1}$ in rural Cedar Creek, which experienced substantial backwater flooding from the seventh-order Kansas River, to $4.34 \text{ m}^3 \text{ s}^{-1}$ in urban Indian Creek. In general, flashiness was greater in urban streams than in rural systems (Figure S2 in Supporting Information S1). Plutonium activity in upland sources ($0.11 \pm 0.07 \text{ Bq kg}^{-1}$) was significantly different, and with large effect size, from subsurface bank sources ($0.04 \pm 0.03 \text{ Bq kg}^{-1}$, $p < 0.01$ and $\delta > 0.50$; Figure 2a). In the transported sediment mixtures, $^{239+240}\text{Pu}$ activities decreased significantly with urbanization from $0.08 \pm 0.02 \text{ Bq kg}^{-1}$ in Blue River to $0.04 \pm 0.01 \text{ Bq kg}^{-1}$ in Indian Creek ($p < 0.01$ and $\delta > 0.50$; Figure 2a), indicating an increase in the contribution of bank sediment with urban land use.

Plutonium sediment fingerprinting results showed increasing streambank contribution with urban land use (Figures 2b and 2c). For rural systems, fingerprinting results show considerable temporal variability in the contribution of bank sediment. In Blue River, for example, Bayesian model estimates of bank sourcing range from a low of 10% (0%–33% at the 95% credible interval) to a high of 82% (47%–98%). On the other hand, urban streams show near-complete dominance of bank sourcing irrespective of time with Mill Creek contributions ranging from a low of 84% (53%–98%) to a high of 96% (83%–100%). The preference for bank sourcing is further reflected in the average Bayesian model results and narrower credible intervals in urban watersheds, ranging from 50% (8%–80%; Blue River) to 93% (73%–100%; Indian Creek, Figure S3 in Supporting Information S1).

Median frequentist modeling results broadly follow the trend of Bayesian predictions although with greater uncertainty and wider credible intervals (Figure 2c). Bayesian credible intervals are 0.64 ± 0.25 the width of frequentist intervals. Frequentist predictions are particularly uncertain in the urban streams where transported sediment is primarily from streambanks. Bayesian and frequentist mixing models applied to the same data set are recognized to provide variable results due to underlying differences in their model structures (Cooper et al., 2014; Li et al., 2020). Greater uncertainty in the frequentist model could be due to the relatively small sample size of the end-members ($n = 11$ for bank and 21 for upland, Figure 2a). Additional sampling of end-members could help to alleviate this issue. While most sediment fingerprinting studies utilize frequentist approaches due, in part, to the ease of implementation (Collins et al., 2020), we recommend adaption of Bayesian approaches as they consider the entire distributions of all parameters together, and yield more realistic distributions (Cooper & Krueger, 2017).

In this study, $^{239+240}\text{Pu}$ showed success for apportioning upland and bank sediment sources. Prior attempts in our study area to apply other radionuclides were unsuccessful (^{137}Cs and $^{210}\text{Pb}_{\text{ex}}$; Lee et al., 2009). In the Lee and others (2009) study, ^{137}Cs mixing model results indicated $106 \pm 22\%$ contribution from streambank soil, falling outside of physical limits. One explanation for these differences could include the time-integrated nature of our sampling design, whereby in situ sediment traps passively fill over the course of 2 weeks (Phillips et al., 2000), rather than point-in-time grab sampling (Lee et al., 2009). Another explanation could be in how the upland/surface signature is classified and quantified. Upland areas like hills could in reality feed subsurface sediment to streams if deep rills and gullies were to erode. While ^{137}Cs is almost entirely bound to surficial sediment, causing an exponential distribution with depth (Wang et al., 2017), $^{239+240}\text{Pu}$ may take a polynomial distribution with depth due to slight downward migration of plutonium (Alewell et al., 2017 and references within). This downward migration causes elevated $^{239+240}\text{Pu}$ activities to persist slightly below the soil surface, preserving the upland soil signature in locations with limited rill and interrill erosion. However, the depth profile of radionuclides can vary considerably due to postdepositional alterations and soil cores should be collected when possible to verify assumptions (Alewell et al., 2017; Ravichandran et al., 1995; Sholkovitz, 1983). The vast majority of applications to date of $^{239+240}\text{Pu}$ activity have been constrained to sediment chronology and mixing studies of deposited material in lakes, hillslopes, and floodplains (e.g., Arata et al., 2016; Calitri et al., 2019; Evrard et al., 2014; Johansen et al., 2019; Ketterer et al., 2004; Smith et al., 2012; Tims et al., 2010; Meusburger et al., 2016). Here, we show a novel application of $^{239+240}\text{Pu}$ for highlighting the likely sources of fluvial sediment transport at short, storm-event timescales.

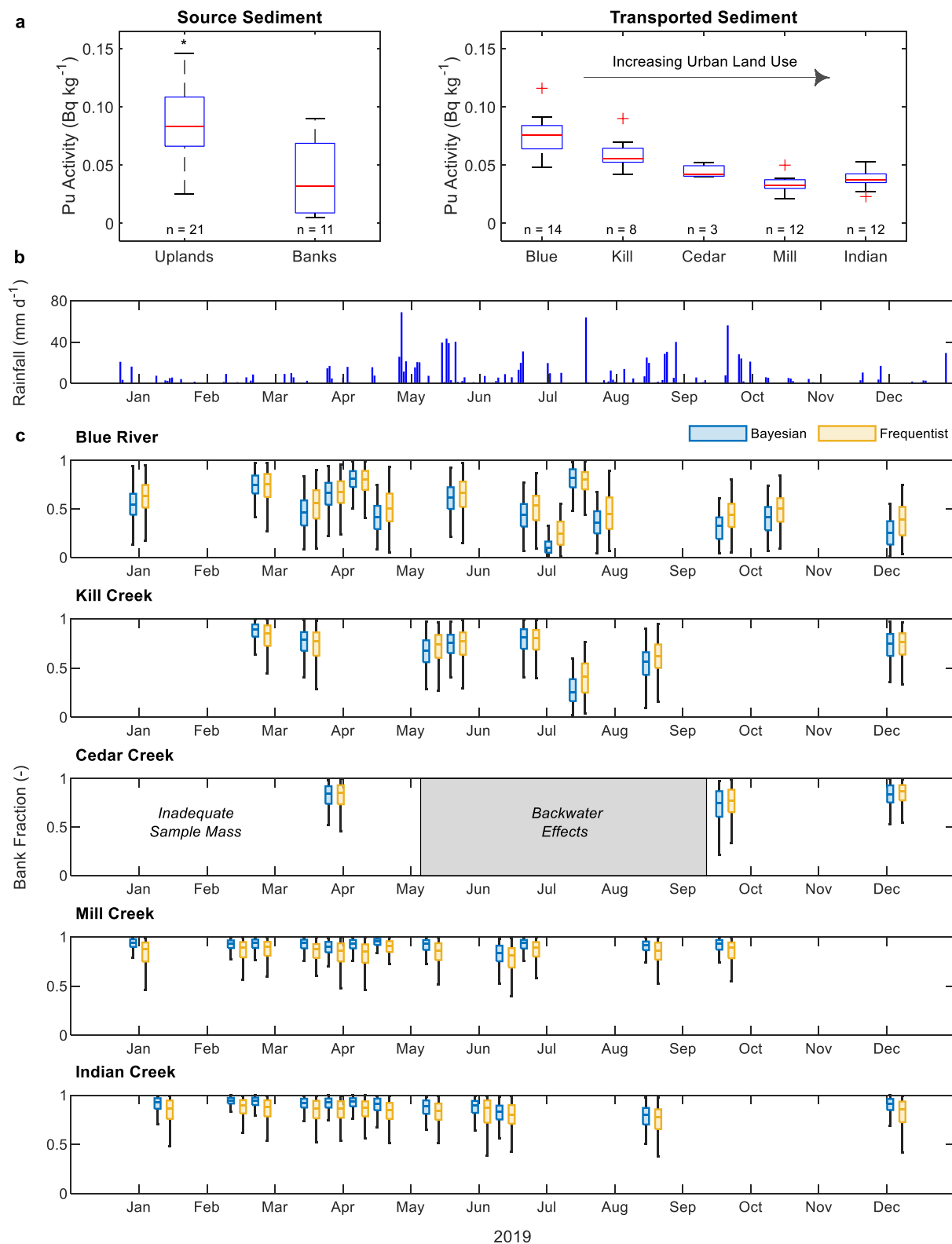


Figure 2. (a) Plutonium activity in source and transported sediment. *Four upland outliers plot outside axis range. (b) Mean precipitation in study area. (c) Bayesian and Frequentist bank contribution estimates with median, 50%, and 95% credible intervals shown on each box-plot.

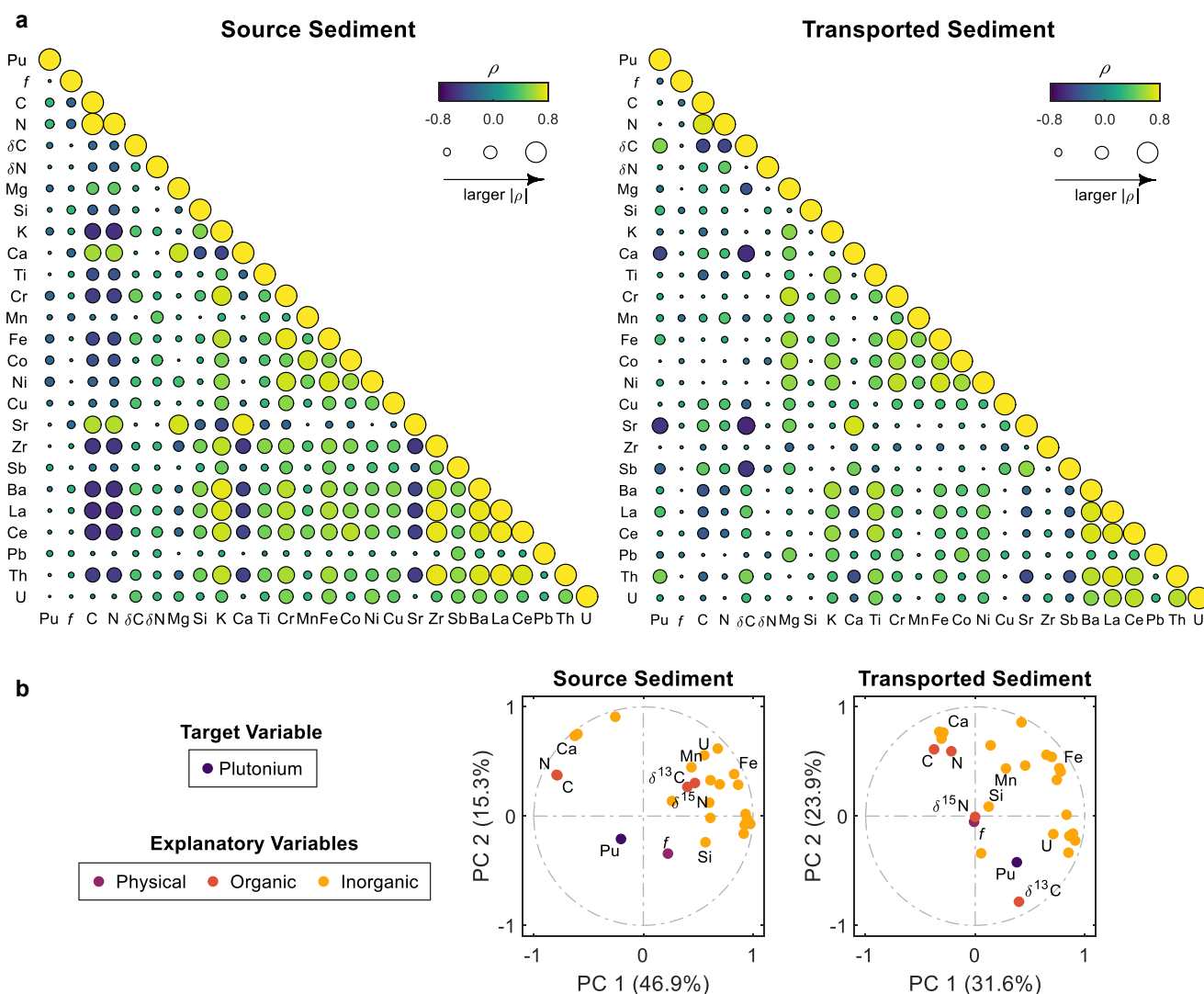


Figure 3. (a) Correlation matrices of Plutonium activity (Pu), fine fraction (f), organic carbon (C) and nitrogen (N), carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopic ratios, and 20 inorganic elements for all source and transported sediment. $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ are abbreviated as δC and δN , respectively, to minimize clutter in plot axes. (b) PCA plots for all source and transported sediment. Note: only a select subset of inorganic elements is labeled for visual clarity.

4.2. Physical and Geochemical Drivers of Plutonium Activity

The majority of transported ($72 \pm 13\%$) and source ($63 \pm 13\%$) material was fine sediment ($d < 63 \mu\text{m}$) with a median particle size (d_{50}) of $14.5 \pm 3.1 \mu\text{m}$, indicating a fine silt composition. No significant particle size effects were observed when comparing $^{239+240}\text{Pu}$ activity and d_{50} , likely due to the narrow range of particle sizes in our samples (Figure S4 in Supporting Information S1). Our correlation analyses showed moderate and strong correlation of $^{239+240}\text{Pu}$ activity with several physical, organic, and inorganic variables in both source and transported sediment (Figure 3a). In source samples, only moderate correlations with $^{239+240}\text{Pu}$ activity exist for C and N ($\rho = 0.33$ – 0.35) and Fe, Co, Ni, and Cr ($\rho = -0.32$ to -0.37). Regarding transported samples, strong correlations were found with $\delta^{13}\text{C}$ ($\rho = 0.63$), Ca and Sr ($\rho = -0.58$ to -0.70), and Th ($\rho = 0.59$) while moderate correlations exist with Si, Fe, and La ($\rho = 0.36$ – 0.42) and Sb ($\rho = -0.42$). Regarding PCA results, the first two principal components explain 56%–62% of the variability in the data set (Figure 3b). In source sediment, $^{239+240}\text{Pu}$ activity is located in a PCA space separate from all explanatory variables. For transported sediment, $^{239+240}\text{Pu}$ activity plots closely with $\delta^{13}\text{C}$ and opposite of C, N, and Ca.

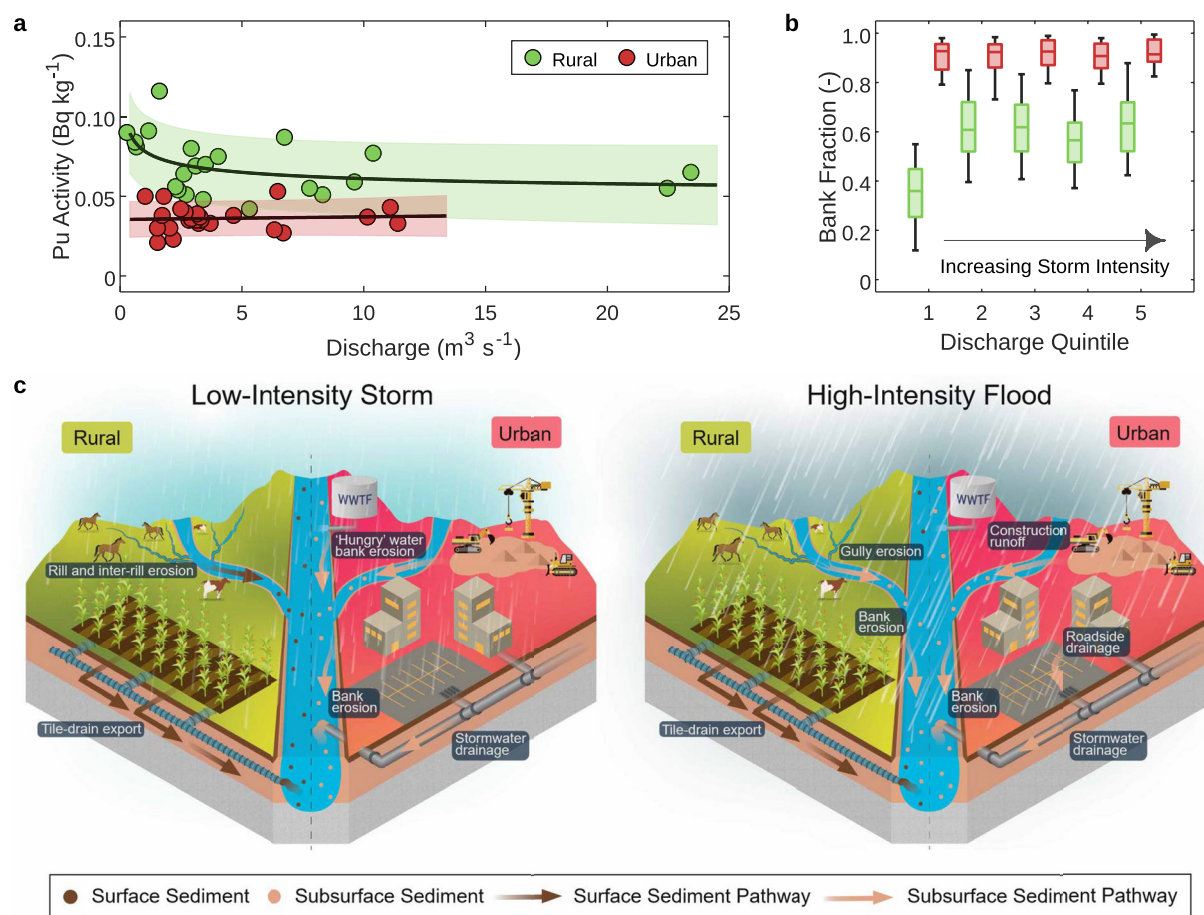


Figure 4. (a) Plutonium activity of rural and urban transported sediment as a function of mean event discharge. Fitted regressions and their associated uncertainty are shown in solid lines and shaded bounds, respectively. (b) Bayesian bank fraction estimates for rural and urban streams grouped by discharge quintile. (c) Conceptual model of storm-dependent sources and pathways of sediment as elucidated by plutonium activity.

Plutonium activity provided unique explanatory power in our urbanizing catchments, separate from that given by organic constituents and inorganic elements. In our source samples (Figure 3), we did not find strong correlation of $^{239+240}\text{Pu}$ activity with any of the other tracers; however, in transported sediment, which are primarily composed of bank material (Figure 2b), strong positive correlation emerged with $\delta^{13}\text{C}$ and negative correlation with Ca/Sr. Negative correlation with Ca/Sr was expected given that the study area is underlain by limestone (i.e., CaCO_3 ; Figure S1 in Supporting Information S1), thus subsurface sediment (lower $^{239+240}\text{Pu}$ activity) are more geogenic (higher Ca). Further, higher $\delta^{13}\text{C}$ composition with lower $^{239+240}\text{Pu}$ activity could also indicate dissolution of carbonates in the subsurface and/or sourcing of depleted organic matter (negative correlation with C and N), which has fractionated. Our results agree with several studies that have shown a dependence of $^{239+240}\text{Pu}$ activity on Fe and Mn oxyhydroxides (Cook et al., 2020), organic matter (Alewell et al., 2017), and particle size (Dong et al., 2017), but these relationships are weak-to-moderate ($|r| < 0.40$), potentially because of the relatively homogenous rock type, soil texture, and particle distribution across our basins. Thus, variability in $^{239+240}\text{Pu}$ activity appears to be driven more by the superposition, that is, vertical position of soil in the landscape, than by other geochemical drivers, although they do impart some effect.

4.3. Urbanization Impacts to Fluvial Sediment Sourcing

Plutonium activity results indicate that sediment sourcing in urbanized watersheds is independent of discharge, whilst sediment sourcing in agricultural watersheds varies with flow intensity (Figure 4a). $^{239+240}\text{Pu}$ activity of transported sediment in urban basins mirrors the bank source at all streamflows. On the other hand, $^{239+240}\text{Pu}$ activity in agricultural stream sediment resembles the upland source at low flows but trends toward the bank

signature as flow increases. Likewise, Bayesian modeling reflects this shift in rural sediment sourcing from 36% (12%–55%) bank contribution at low discharges to 63% (42%–88%) at high flows (Figure 4b).

Plutonium fingerprinting allows for a conceptual model of land use impacts on the sources and pathways of sediment during low- and high-intensity events (Figure 4c). For low-intensity storms in rural watersheds, dominant pathways include rill and inter-rill erosion as well as tile-drainage export of surficial sediment from cropland. Tile-drains can artificially augment the connectivity of distal sediment by piping upland material directly to stream corridors (Nazari et al., 2020; Williams et al., 2016). This pathway is plausible as tile-drained area in the rural watersheds ranges from 12.2% to 18.6% (Figure 1). On the other hand, in urban watersheds, low-intensity events fall on impervious surfaces, which deliver high-energy sediment-free flows to ditches and tributaries where substantial bank erosion occurs (Russell et al., 2019; Zarnaghsh & Husic, 2021). Further, wastewater treatment facilities discharge sediment-starved water, which have a high capacity for erosion, directly into streams (Lawler et al., 2006). This is plausible as facility outflows can account for up to 80% of Indian Creek discharge during baseflow (Zarnaghsh & Husic, 2021). On the other hand, at high flows, surficial rill and inter-rill erosion transition to subsurface gully and bank erosion in rural basins (Shi et al., 2021) whereas subsurface sediment erosion continues to prevail in urban basins. In some ways, a year of record flooding allows for the connectivity of sediment at unprecedented levels so the results presented here may not be entirely representative of a typical hydrologic year. As urban development and agricultural activity increases to meet growing population demands, decreasing sedimentation in fluvial systems will be integral to creating sustainable and efficient urban and agricultural systems. This is particularly important as subsurface erosion is likely to increase in the future with projected rises in runoff due to land use and climate change (Rode et al., 2018).

5. Conclusions

We successfully demonstrate the utility of $^{239+240}\text{Pu}$ activity as a tracer of fluvial sediment in urbanizing catchments. $^{239+240}\text{Pu}$ activity decreased as catchments urbanized, indicating a greater contribution of subsurface bank sediment, ranging from 50% (8%–80%) in the most rural watershed and up to 93% (73%–100%) in the most urban catchment. Plutonium activities also correctly classify rills and interrills as upland erosion sources, which may result from downward migration of Pu in soil profiles at source areas. We also find that $^{239+240}\text{Pu}$ activity provides explanatory information on the superposition of sediment sources, which is beyond that given by traditional organic and geochemical tracers that primarily infer vegetative and geologic sourcing, respectively.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

Open Research: Data, model results, and code will be stored and publicly available on Open Science Framework (at: <http://www.doi.org/10.17605/OSF.IO/NXDQW>).

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